

Masked Wheel Generation Regression

Date: 2026-06-02

Test case: `ivan-C1`

Branch Scope

This branch started as a Stage 2 mask-quality experiment:

`detected wheel candidates -> VLM keep/reject filter -> cleaner wheel mask`

After the mask filter looked useful, the scope expanded to a generation eval:

`source car + target wheel reference + generated/filtered mask
-> wheel replacement model`

That means the branch now contains two related but separate questions:

1. Does the VLM filter improve the wheel mask?
2. If we pass that mask into generation/inpaint models, which model produces the best wheel replacement?

The pages below answer the second question for the current corrected rerun.

Corrected Rerun

The previous frontier page included outputs generated with an old prompt that asked for a matte-black wheel while the actual reference image was silver. That made several rows visually misleading. The rerun below uses the corrected silver prompt:

`Use the exact wheel design, color, finish, spoke pattern, center cap, and material from the reference image.`

`Preserve original scene lighting.`

`Do not alter car body, car color, background, road, tires, or unmasked pixels.`

Cases:

- `ivan-C1`: daylight control case.
- `ivan-N1`: night case.
- `ivan-N2`: night wet-looking case. There is no explicit rain metadata in the local dataset, so this is not labeled as confirmed rain.

01. ivan-C1 (daylight control) / AITUNNEL gpt-image-1.5

AITUNNEL / OpenAI-compatible edits | corrected silver prompt | ref passed: yes | mask passed: yes

Input image: source car



Input image: wheel reference



Output / status



02. ivan-C1 (daylight control) / Gemini 3 Pro Image Edit corrected

fal.ai / gemini-3-pro-image-preview/edit | corrected silver prompt | ref passed: yes | mask passed: yes

Input image: source car



Input image: wheel reference



Output / status



03. ivan-C1 (daylight control) / Reve Direct masked corrected

Reve API / image remix | corrected silver prompt | ref passed: yes | mask passed: yes

Input image: source car



Input image: wheel reference



Output / status



04. ivan-N1 (night) / AITUNNEL gpt-image-1.5

AITUNNEL / OpenAI-compatible edits | corrected silver prompt | ref passed: yes | mask passed: yes

Input image: source car



Input image: wheel reference



Output / status



05. ivan-N1 (night) / Gemini 3 Pro Image Edit corrected

fal.ai / gemini-3-pro-image-preview/edit | corrected silver prompt | ref passed: yes | mask passed: yes

Input image: source car



Input image: wheel reference



Output / status



06. ivan-N1 (night) / Reve Direct masked corrected

Reve API / image remix | corrected silver prompt | ref passed: yes | mask passed: yes

Input image: source car



Input image: wheel reference



Output / status



07. ivan-N2 (night wet-looking) / AITUNNEL gpt-image-1.5

AITUNNEL / OpenAI-compatible edits | corrected silver prompt | ref passed: yes | mask passed: yes

Input image: source car



Input image: wheel reference



Output / status



08. ivan-N2 (night wet-looking) / Gemini 3 Pro Image Edit corrected

fal.ai / gemini-3-pro-image-preview/edit | corrected silver prompt | ref passed: yes | mask passed: yes

Input image: source car



Input image: wheel reference



Output / status



09. ivan-N2 (night wet-looking) / Reve Direct masked corrected

Reve API / image remix | corrected silver prompt | ref passed: yes | mask passed: yes

Input image: source car



Input image: wheel reference



Output / status



Corrected rerun status:

Model	Cases	Status	Read
gpt-image-1.5 via AITUNNEL	C1, N1, N2	3/3 completed after one N2 retry	best current OpenAI-compatible fallback through AITUNNEL
Gemini 3 Pro Image Edit via fal.ai	C1, N1, N2	3/3 completed	no longer black, but often fails the edit task by cropping or generating product-wheel style outputs
Reve Direct masked corrected	C1, N1, N2	3/3 completed	stable masked baseline

gpt-image-1.5 note: the first N2 request failed with `Server disconnected without sending a response`; a single retry completed. This means AITUNNEL edits are usable but still have transport-level instability.

Rain Stress Cases

Additional user-provided images were tested with the same corrected silver prompt and manually prepared wheel masks.

Cases:

- `rain-flood-bmw`: flooded road / water splash.
- `night-heavy-rain-sedan`: night heavy rain.
- `istock-rain-splash`: rain splash with visible stock watermark; this is a stress-test only, not a presentation-quality source.

01. rain-flood-bmw (flooded road / water splash) / AITUNNEL gpt-image-1.5

AITUNNEL / OpenAI-compatible edits | corrected silver prompt | ref passed: yes | mask passed: yes

Input image: source car



Input image: wheel reference



Output / status



02. rain-flood-bmw (flooded road / water splash) / Gemini 3 Pro Image Edit corrected

fal.ai / gemini-3-pro-image-preview/edit | corrected silver prompt | ref passed: yes | mask passed: yes

Input image: source car



Input image: wheel reference



Output / status



03. rain-flood-bmw (flooded road / water splash) / Reve Direct masked corrected

Reve API / image remix | corrected silver prompt | ref passed: yes | mask passed: yes

Input image: source car



Input image: wheel reference



Output / status



04. night-heavy-rain-sedan (night heavy rain) / AITUNNEL gpt-image-1.5

AITUNNEL / OpenAI-compatible edits | corrected silver prompt | ref passed: yes | mask passed: yes

Input image: source car



Input image: wheel reference



Output / status

FAILED

failed twice: Server disconnected without sending a response.

05. night-heavy-rain-sedan (night heavy rain) / Gemini 3 Pro Image Edit corrected

fal.ai / gemini-3-pro-image-preview/edit | corrected silver prompt | ref passed: yes | mask passed: yes

Input image: source car



Input image: wheel reference



Output / status



06. night-heavy-rain-sedan (night heavy rain) / Reve Direct masked corrected

Reve API / image remix | corrected silver prompt | ref passed: yes | mask passed: yes

Input image: source car



Input image: wheel reference



Output / status



07. istock-rain-splash (rain splash / watermark stress-case) / AITUNNEL gpt-image-1.5

AITUNNEL / OpenAI-compatible edits | corrected silver prompt | ref passed: yes | mask passed: yes

Input image: source car



Input image: wheel reference



Output / status



08. istock-rain-splash (rain splash / watermark stress-case) / Gemini 3 Pro Image Edit corrected

fal.ai / gemini-3-pro-image-preview/edit | corrected silver prompt | ref passed: yes | mask passed: yes

Input image: source car



Input image: wheel reference



Output / status



09. istock-rain-splash (rain splash / watermark stress-case) / Reve Direct masked corrected

Reve API / image remix | corrected silver prompt | ref passed: yes | mask passed: yes

Input image: source car



Input image: wheel reference



Output / status



Rain stress status:

Model	Cases	Status	Read
gpt-image-1.5 via AITUNNEL	rain-flood-bmw, night-heavy-rain-sedan, istock-rain-splash	2/3 completed; night-heavy-rain-sedan failed twice	useful when it completes, but transport instability remains
Gemini 3 Pro Image Edit via fal.ai	all 3 rain cases	3/3 completed	still inconsistent as masked car editor
Reve Direct masked corrected	all 3 rain cases	3/3 completed	most stable under these stress cases

Status Summary

Current corrected rerun:

Model	Cases	Status	Result
gpt-image-1.5 via AITUNNEL	C1, N1, N2	completed after one N2 retry	usable OpenAI-compatible fallback
Gemini 3 Pro Image Edit via fal.ai	C1, N1, N2	completed	still not reliable for masked car edits
Reve Direct masked	C1, N1, N2	completed	stable masked baseline
gpt-image-1.5 via AITUNNEL	rain stress cases	2/3 completed; one case failed twice	transport instability remains
Gemini 3 Pro Image Edit via fal.ai	rain stress cases	completed	visually inconsistent
Reve Direct masked	rain stress cases	completed	stable

AITUNNEL endpoint diagnostics:

Model	Endpoint	Status
gpt-image-2	/v1/images/generations	works
gpt-image-2	/v1/images/edits	fails: Server disconnected without sending a response.
gpt-image-1.5	/v1/images/edits with image[] + image[] + mask	works, but one transient disconnect on N2
gpt-image-1-mini	/v1/images/edits with image[] + image[] + mask	works
gpt-image-1	/v1/images/edits	fails: Server disconnected without sending a response.

Conclusion:

- AITUNNEL now supports the required image[] + mask request shape for gpt-image-1.5 and gpt-image-1-mini.
- gpt-image-2 still fails on /v1/images/edits, including simpler edit control requests.
- For the current wheel-copy eval, gpt-image-1.5 is the usable AITUNNEL model.

GPT Image 2 Decision

gpt-image-2 remains the main candidate to test as a potential Reve replacement because it should support the exact request shape required by this product:

source car + target wheel reference + alpha mask

Current AITUNNEL status:

- gpt-image-2 is visible in /v1/models.
- gpt-image-2 works for /v1/images/generations.
- gpt-image-2 still fails for /v1/images/edits.
- The failure is transport-level disconnect, not a JSON validation error.

Operational decision:

- Do not block the report on gpt-image-2.
- Use AITUNNEL gpt-image-1.5 as the current OpenAI-compatible fallback for demos and regression checks.
- Do not call gpt-image-1.5 a full Reve replacement yet; it needs more cases and stability checks.
- Keep asking AITUNNEL support to fix /v1/images/edits for gpt-image-2.

Next Steps

- Run the fine-tuned real/fake wheel classification model on generated outputs to separate true wheel-copy results from visually plausible but incorrect wheel generations.
- Improve wheel masking quality before the next model comparison; current manual and filtered masks are good enough for stress testing, but mask edge quality still affects rain, splash, and night cases.

Output Files

- Corrected rerun summary CSV: docs/assets/aitunnel-gpt-image-regression/corrected-rerun-c1-n1-n2-summary.csv
- Rain stress summary CSV: docs/assets/aitunnel-gpt-image-regression/rain-rerun-summary.csv

Raw Results

- tmp/openai-image-edit-eval/results-aitunnel-gpt-image-15-corrected-c1-n1-n2/openai_image_edit_results.csv
- tmp/openai-image-edit-eval/results-aitunnel-gpt-image-15-corrected-n2-retry/openai_image_edit_results.csv
- tmp/fal-inpaint-eval/results-corrected-silver-gemini-c1-n1-n2/fal_inpaint_results.csv
- tmp/reve-image-edit-eval/results-corrected-silver-masked-c1-n1-n2/reve_image_edit_results.csv
- tmp/rain-wheel-eval/results-aitunnel-gpt-image-15/openai_image_edit_results.csv
- tmp/rain-wheel-eval/results-aitunnel-gpt-image-15-night-heavy-rain-retry/openai_image_edit_results.csv
- tmp/rain-wheel-eval/results-gemini/fal_inpaint_results.csv
- tmp/rain-wheel-eval/results-reve-masked/reve_image_edit_results.csv